

RaceDay SQL Database Script

PROG6212 - Programming 2B | PoE Part 1 | Section C (20 Marks) — Readable PDF copy. The runnable file is RaceDay_Database.sql

This document is a print/read copy of the SQL Server script for reference and for the video walkthrough. The authoritative, executable file that must live in /docs and be run in SSMS is **RaceDay_Database.sql**.

```
/* =====
RaceDay Event Management System
SQL Server Database Script - PROG6212 PoE Part 1
=====
Run on a clean SQL Server instance in SSMS.
Creates: Role, [User], Event, Category, Enrolment, Result
===== */

IF DB_ID('RaceDayDB') IS NULL
BEGIN
    CREATE DATABASE RaceDayDB;
END
GO

USE RaceDayDB;
GO

/* ----- Drop tables if they already exist (clean rebuild) ----- */
IF OBJECT_ID('dbo.Result', 'U') IS NOT NULL DROP TABLE dbo.Result;
IF OBJECT_ID('dbo.Enrolment', 'U') IS NOT NULL DROP TABLE dbo.Enrolment;
IF OBJECT_ID('dbo.Category', 'U') IS NOT NULL DROP TABLE dbo.Category;
IF OBJECT_ID('dbo.Event', 'U') IS NOT NULL DROP TABLE dbo.Event;
IF OBJECT_ID('dbo.[User]', 'U') IS NOT NULL DROP TABLE dbo.[User];
IF OBJECT_ID('dbo.Role', 'U') IS NOT NULL DROP TABLE dbo.Role;
GO

/* =====
1. Role (lookup table: Organiser / Participant)
===== */
CREATE TABLE dbo.Role (
    RoleID INT IDENTITY(1,1) PRIMARY KEY,
    RoleName VARCHAR(20) NOT NULL UNIQUE
);
GO

/* =====
2. User (both Organisers and Participants, distinguished by RoleID)
===== */
CREATE TABLE dbo.[User] (
    UserID INT IDENTITY(1,1) PRIMARY KEY,
    FullName VARCHAR(100) NOT NULL,
    Email VARCHAR(150) NOT NULL UNIQUE,
    PasswordHash VARCHAR(255) NOT NULL,
    ContactNumber VARCHAR(20) NULL,
    EmergencyContact VARCHAR(20) NULL,
    RoleID INT NOT NULL,
    CreatedDate DATETIME NOT NULL DEFAULT GETDATE(),
    CONSTRAINT FK_User_Role FOREIGN KEY (RoleID) REFERENCES dbo.Role(RoleID)
);
GO

/* =====
3. Event (created and owned by an Organiser)
===== */
CREATE TABLE dbo.Event (
    EventID INT IDENTITY(1,1) PRIMARY KEY,
    EventName VARCHAR(150) NOT NULL,
    Description VARCHAR(500) NULL,
    EventDate DATE NOT NULL,
    Location VARCHAR(150) NOT NULL,
    DistanceKm DECIMAL(5,2) NOT NULL,
    EventType VARCHAR(20) NOT NULL CHECK (EventType IN ('Running','Walking','Cycling')),
    MaxParticipants INT NOT NULL DEFAULT 100,
    Status VARCHAR(20) NOT NULL DEFAULT 'Published'
    CHECK (Status IN ('Draft','Published','Cancelled','Completed')),
    OrganiserID INT NOT NULL,
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        CONSTRAINT FK_Event_Organiser FOREIGN KEY (OrganiserID) REFERENCES dbo.[User](UserID)
    );
GO

/* =====
4. Category (belongs to one Event)
===== */
CREATE TABLE dbo.Category (
    CategoryID INT IDENTITY(1,1) PRIMARY KEY,
    CategoryName VARCHAR(50) NOT NULL,
    MinAge INT NOT NULL DEFAULT 0,
    MaxAge INT NOT NULL DEFAULT 99,
    Fee DECIMAL(8,2) NOT NULL DEFAULT 0,
    EventID INT NOT NULL,
    CONSTRAINT FK_Category_Event FOREIGN KEY (EventID) REFERENCES dbo.Event(EventID)
);
GO

/* =====
5. Enrolment (associative entity resolving the Participant <-> Event
<-> Category many-to-many relationships)
===== */
CREATE TABLE dbo.Enrolment (
    EnrolmentID INT IDENTITY(1,1) PRIMARY KEY,
    EnrolmentDate DATETIME NOT NULL DEFAULT GETDATE(),
    Status VARCHAR(20) NOT NULL DEFAULT 'Confirmed'
    CHECK (Status IN ('Pending', 'Confirmed', 'Cancelled')),
    ParticipantID INT NOT NULL,
    EventID INT NOT NULL,
    CategoryID INT NOT NULL,
    CONSTRAINT FK_Enrolment_Participant FOREIGN KEY (ParticipantID) REFERENCES dbo.[User](UserID),
    CONSTRAINT FK_Enrolment_Event FOREIGN KEY (EventID) REFERENCES dbo.Event(EventID),
    CONSTRAINT FK_Enrolment_Category FOREIGN KEY (CategoryID) REFERENCES dbo.Category(CategoryID),
    CONSTRAINT UQ_Enrolment_ParticipantEvent UNIQUE (ParticipantID, EventID)
);
GO

/* =====
6. Result (one-to-one with Enrolment; captured by an Organiser)
===== */
CREATE TABLE dbo.Result (
    ResultID INT IDENTITY(1,1) PRIMARY KEY,
    FinishTime TIME NULL,
    FinishPosition INT NULL,
    CapturedDate DATETIME NOT NULL DEFAULT GETDATE(),
    EnrolmentID INT NOT NULL UNIQUE,
    CapturedByOrganiserID INT NOT NULL,
    CONSTRAINT FK_Result_Enrolment FOREIGN KEY (EnrolmentID) REFERENCES dbo.Enrolment(EnrolmentID),
    CONSTRAINT FK_Result_Organiser FOREIGN KEY (CapturedByOrganiserID) REFERENCES dbo.[User](UserID)
);
GO

/* =====
SEED DATA
===== */

-- Roles
INSERT INTO dbo.Role (RoleName) VALUES ('Organiser'), ('Participant');
GO

-- 2 Organisers + 2 Participants
INSERT INTO dbo.[User] (FullName, Email, PasswordHash, ContactNumber, EmergencyContact, RoleID)
VALUES
('Thabo Nkosi', 'thabo.nkosi@raceday.co.za', 'HASHED_PWD_1', '0821234567', NULL, 1),
('Lindiwe Dube', 'lindiwe.dube@raceday.co.za', 'HASHED_PWD_2', '0837654321', NULL, 1),
('Sipho Mokoena', 'sipho.mokoena@example.com', 'HASHED_PWD_3', '0731122334', '0827654321', 2),
('Amahle Botha', 'amahle.botha@example.com', 'HASHED_PWD_4', '0765544332', '0839988776', 2);
GO

-- 3 Events (owned by the two Organisers)
INSERT INTO dbo.Event (EventName, Description, EventDate, Location, DistanceKm, EventType, MaxParticipants, Status, Orga
VALUES
('Johannesburg City 10K', 'Annual road running race through the Johannesburg CBD.', '2026-10-18', 'Johannesburg, Gauteng', 10, 'Road Race', 100, 'Open', 1),
('Cape Town Cycle Classic', 'Scenic cycling event along the Cape Peninsula.', '2026-11-08', 'Cape Town, Western Cape', 100, 'Cycling', 50, 'Open', 1),
('Durban Beachfront Walk', 'Family-friendly walking event along the Durban beachfront.', '2026-09-27', 'Durban, KwaZulu-Natal', 5, 'Walking', 500, 'Open', 2);
GO

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-- Categories for each Event
INSERT INTO dbo.Category (CategoryName, MinAge, MaxAge, Fee, EventID)
VALUES
('Open Race', 18, 65, 150.00, 1),
('Junior Race', 10, 17, 80.00, 1),
('Seeded Riders', 18, 70, 450.00, 2),
('Fun Ride', 12, 99, 250.00, 2),
('Family Walk', 5, 99, 50.00, 3);
GO

-- Sample Enrolments (Participants entering Events/Categories)
INSERT INTO dbo.Enrolment (Status, ParticipantID, EventID, CategoryID)
VALUES
('Confirmed', 3, 1, 1),
('Confirmed', 4, 1, 1),
('Confirmed', 3, 2, 3),
('Pending', 4, 3, 5);
GO

-- Sample Results (captured by an Organiser for a completed enrolment)
INSERT INTO dbo.Result (FinishTime, FinishPosition, EnrolmentID, CapturedByOrganiserID)
VALUES
('00:42:13', 5, 1, 1),
('00:45:02', 12, 2, 1);
GO

```